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The Allocation of Talent and U.S. Economic Growth

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1 Big picture

- **Question.** How much U.S. growth from 1960 to 2010 can be explained by a better allocation of talent across occupations as white women, black men, and black women entered occupations that had previously been dominated by white men?
- **Motivating fact.** In 1960, doctors and lawyers were overwhelmingly white men; by 2010, the representation of women and black men in these and other high-skilled occupations had risen sharply. The paper interprets this convergence as evidence that many talented workers were previously prevented, discouraged, or distorted away from their comparative advantage.
- **Model lens.** The paper uses a general-equilibrium Roy model of occupational choice. People differ in occupation-specific talent or preferences. They choose occupations after taking into account wages, human capital costs, labor-market wedges, and occupation-specific preferences or social norms.
- **Three barriers.** The model allows: (i) labor market discrimination, modeled as a wedge between the wage and marginal product; (ii) barriers to human capital accumulation, modeled as higher monetary costs of acquiring occupation-specific human capital; and (iii) group-specific occupational preferences or social norms, including norms about market work versus home production.
- **Main baseline result.** Falling barriers facing women and black men explain about 41.5% of growth in market GDP per person from 1960 to 2010 in the baseline case where workers sort on occupation-specific talent.
- **Most important source.** Declining human capital barriers are quantitatively more important than declining labor market discrimination. In the baseline decomposition, human capital barriers account for 36.0% of growth in market GDP per person, while labor market barriers account for 7.7%.
- **Economic takeaway.** Growth can come not only from technology and capital accumulation, but also from reducing misallocation of people across jobs. A large part of U.S. growth over the last half-century reflects allowing talented people to enter high-productivity occupations.

2 Incremental contribution of the paper

- **Quantifies a macroeconomic cost of occupational exclusion.** Existing work documented gender and racial convergence in labor-market outcomes, but this paper translates that convergence into aggregate growth using a structural model.
- **Connects discrimination and growth.** The paper treats discrimination and barriers to skill acquisition as allocative distortions, analogous to misallocation in production, but applied to people and occupations.
- **Uses occupational shares, not just wage gaps.** A central conceptual point is that wage gaps are not robust measures of barriers because selection can offset wage effects. Occupational quantities are more informative about frictions.
- **Distinguishes human capital barriers from labor market barriers.** The life-cycle structure lets the authors treat pre-market barriers as cohort effects and labor market discrimination as time effects.

- **Incorporates the home sector.** The model treats home production as another occupation, allowing the authors to connect falling occupational barriers to rising labor force participation.
- **Provides a robustness bridge between talent and preference sorting.** The baseline assumes workers sort on talents; robustness exercises allow sorting on idiosyncratic preferences. Even under preference sorting, the paper finds meaningful growth effects.
- **General-equilibrium accounting.** The model captures how the movement of women and black men into high-skilled occupations affects wages of white men and the price of skills across occupations, not only the earnings of entrants.

3 Core mechanism and economic intuition

3.1 Misallocation of talent

The basic idea is that each person has a comparative advantage across occupations. If barriers prevent people from choosing the occupation where their talent is highest, aggregate output falls. When those barriers decline, talented workers reallocate toward occupations where their skills have high social value.

The mechanism is especially powerful when the excluded groups are large. White women are a large share of the population, so barriers that prevent talented women from entering high-skilled market occupations can have large aggregate effects. Black men and black women are smaller population shares, so their aggregate contribution is smaller even when their individual barriers are large.

3.2 Three distortions

- **Labor market discrimination τ_{ig}^w .** Workers in group g and occupation i receive less take-home compensation per efficiency unit than their marginal product. This can represent employer discrimination, customer discrimination, or other wage wedges.
- **Human capital barriers τ_{ig}^h .** Workers face higher costs of acquiring occupation-specific human capital. This can capture unequal schooling, barriers to college or professional training, differential parental investment, wealth constraints, or pre-market socialization.
- **Occupational preferences/social norms $\tilde{z}_{ig}$.** Groups may derive different utility from occupations or from market work relative to home production. The paper treats this flexibly, while emphasizing that preference changes by themselves explain little market GDP growth in the baseline model.

The composite barrier is

$$\tau_{ig} = \frac{(1 + \tau_{ig}^h)^\eta}{1 - \tau_{ig}^w}. \quad (1)$$

A larger τ_{ig} lowers the utility-adjusted payoff of occupation i for group g , and therefore lowers that group's occupational share in that occupation.

3.3 Why quantities are more informative than wage gaps

One of the paper's most important conceptual points is that wage gaps are not enough to infer barriers. Suppose women face a high barrier in becoming lawyers. If only the most talented women overcome the barrier, the average wage of women lawyers may not be low. In fact, strong positive selection can make observed wages look relatively high. Thus, wage gaps can understate barriers.

Occupational shares are more robust. If women are rare among lawyers even after controlling for wage gaps, the model reads this as evidence of a high barrier, low relative talent, or low relative occupational preference. Under the baseline normalization of equal relative talent, the residual is attributed to barriers and preferences.

4 Model structure

4.1 Agents, timing, and choices

Each worker belongs to a group g and a cohort c . The four groups are white men, white women, black men, and black women. People choose an occupation i and invest in occupation-specific human capital before entering the labor market. They then work over three life-cycle periods: young, middle-aged, and old. Human capital and occupation are fixed after the initial choice.

Lifetime utility combines consumption, leisure during the pre-market human capital period, group-specific occupational preferences, and idiosyncratic preferences:

$$\log U = \beta \sum_{t=c}^{c+2} \log C(c, t) + \log(1 - s(c)) + \log z_{ig}(c) + \log \mu_i. \quad (2)$$

Human capital is produced using time and goods:

$$h_{ig}(c, t) = \bar{h}_{ig} \gamma(t - c) s_i(c)^{\phi_i} e_{ig}(c)^\eta. \quad (3)$$

Here ϕ_i governs the return to time invested in skill formation, and η governs the elasticity with respect to goods investment.

4.2 Occupational choice

After solving the individual problem, the fraction of people in group g and cohort c choosing occupation i is

$$p_{ig}(c) = \frac{\tilde{w}_{ig}(c)^\theta}{\sum_s \tilde{w}_{sg}(c)^\theta}, \quad (4)$$

where $\tilde{w}_{ig}$ summarizes wages, human capital returns, group talent, preferences, and frictions. The parameter θ governs the dispersion of occupation-specific talent or preferences. A higher θ means less dispersion in comparative advantage.

Interpretation. Occupations attract workers when they have high wage per efficiency unit, high group talent, low human capital barriers, low labor wedges, or favorable group preferences. Occupational convergence over time therefore requires changes in some combination of barriers, preferences, and technology.

4.3 Worker quality and selection

If people sort on ability, average worker quality in an occupation depends on who enters. When a group faces high barriers, only the strongest members enter. When barriers fall, additional marginal workers enter. This selection effect is crucial for understanding why observed wage gaps are not a clean measure of discrimination.

If people sort on preferences instead, occupational shares still change, but selection on ability is absent. The paper studies both cases. The baseline is $\delta = 0$, where all idiosyncratic heterogeneity is in talent. The polar robustness case is $\delta = 1$, where heterogeneity is only in preferences.

4.4 Production side

Final output is produced from occupation-specific efficiency units of labor:

$$Y = \left(\sum_i (A_i H_i)^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}, \quad (5)$$

where A_i is occupation-specific productivity and σ is the elasticity of substitution across occupations. This general equilibrium step matters because the entry of women and black men into high-skilled occupations changes the supply of skills, wages, and the allocation of white men. The model therefore captures effects on all groups, not just the groups whose barriers are falling.

5 Data and measurement

5.1 Data sources

- **Main data.** Decennial U.S. Censuses from 1960, 1970, 1980, 1990, and 2000, plus the pooled 2010–2012 American Community Survey.
- **Groups.** White men, white women, black men, and black women.
- **Ages.** Individuals aged 25–54, divided into young (25–34), middle-aged (35–44), and old (45–54).
- **Occupations.** The market sector has 66 market occupations. The home sector is treated as an additional occupation.
- **Labor force participation.** A person is in the home sector if not employed or working fewer than 10 hours per week. A person working more than 30 hours per week is assigned to a market occupation. Part-time workers are split half to the home sector and half to their reported occupation.
- **Earnings.** Earnings are labor, business, and farm income in the previous year, converted to constant dollars. The earnings sample requires current work, at least 48 weeks worked in the previous year, and at least \$1,000 in earnings in 2010 dollars.

5.2 Synthetic cohorts

The paper follows synthetic cohorts through the life cycle. For example, the cohort that is young in 1960 is middle-aged in 1970 and old in 1980. This cohort structure is what allows the paper to distinguish human capital barriers from labor market discrimination.

5.3 Key empirical objects

- **Occupational propensities.** For each group, occupation, age, and year, the paper measures the probability of working in each occupation relative to white men.
- **Wage gaps.** For each group and occupation, the paper measures log earnings relative to white men.
- **Labor force participation gaps.** The home sector allows the authors to match the movement of women into market work and the changing relative labor force participation of black men.
- **Education by occupation.** Education of young white men disciplines the occupation-specific return to time invested in human capital, ϕ_i .

6 Identification and inference

6.1 Key identifying assumption

The central identifying assumption is that relative mean innate talent across groups in each occupation is constant over time. The baseline goes further and normalizes relative talent to one for all group-occupation cells:

$$\bar{h}_{ig}/\bar{h}_{i,wm} = 1. \quad (6)$$

Under this assumption, changes in occupational sorting and wage gaps are interpreted as changes in barriers, preferences, or technology, not changes in innate group talent.

6.2 Inferring preferences and composite barriers

In the baseline case where workers sort on talent, occupational wage gaps among the young discipline group-specific occupational preferences. The logic is: if a group dislikes an occupation, the group must be compensated to work there, so wages among entrants should be higher.

After preferences are inferred from wage gaps, occupational shares identify composite frictions. If a group is underrepresented in an occupation after accounting for wages and preferences, the model infers a high barrier τ_{ig} .

6.3 Separating τ^h from τ^w

The model separates human capital barriers from labor market discrimination using cohort versus time variation.

- **Human capital barriers are cohort effects.** They affect human capital choices before labor market entry and remain with a cohort over its life cycle.
- **Labor market barriers are time effects.** They affect all cohorts working in a given year.
- **Empirical implementation.** Changes in wage gaps over a cohort's life cycle identify changes in labor-market wedges; differences in intercepts across young cohorts identify pre-market human capital barriers.

For white women, wage-gap profiles mostly shift up across successive cohorts rather than steepen within cohorts. This points to a dominant role for falling human capital barriers. For black men, the data show both intercept shifts and steeper life-cycle profiles, indicating a role for both types of barriers.

6.4 Calibration

The baseline parameters are:

- $\theta = 2$, a compromise between estimates from wage dispersion and labor supply elasticities;
- $\eta = 0.103$, based on education spending relative to output;
- $\sigma = 3$, the baseline elasticity of substitution across occupations;
- $\beta = 0.231$, pinned down by Mincerian returns to schooling.

7 Main results

7.1 Occupational convergence

The dispersion of relative occupational shares fell substantially from 1960 to 2010. This means that the occupational distribution of women and black men became more similar to that of white men. The convergence is strongest for white women and black women in high-skilled occupations, and the model interprets this as evidence that barriers fell sharply.

7.2 Composite frictions declined

The estimated mean composite barrier τ fell sharply for white women, black men, and black women. For white women, the mean fell from around 7 in 1960 to around 3 in 2010. For black women it fell from around 8 to around 3. For black men it fell from around 3 to around 1.5.

The dispersion of $\log \tau$ also fell, meaning that barriers became less occupation-specific. The largest declines were in high-skilled professions, which is why the growth effects are large.

7.3 Preferences changed, but explain little GDP growth

The model infers that white women's average preference for market work relative to the home sector rose over time. However, in the baseline, changing preferences explain little of the growth in market GDP per person. The reason is that if women were simply reluctant to enter certain occupations in 1960, the model would predict compensating wage patterns that are not present in the data.

7.4 Magnitude of growth effects

The baseline counterfactual holds occupational barriers fixed at their 1960 levels while letting technology, returns to schooling, preferences, and group population shares evolve. The gap between actual model growth and this counterfactual is attributed to falling barriers.

- Falling τ^h and τ^w together explain 41.5% of growth in market GDP per person.

- They explain 38.4% of growth in market earnings per person.
- They explain 90.4% of the rise in labor force participation.
- They explain 24.0% of growth in market GDP per worker.
- They explain 32.7% of growth in total GDP per person, including home production.

7.5 Human capital barriers dominate labor market discrimination

Human capital barriers alone account for 36.0% of market GDP per person growth. Labor market discrimination alone accounts for 7.7%. This does not mean labor discrimination is unimportant for earnings; it contributes more to market earnings and labor force participation. But for aggregate output, the pre-market allocation of human capital is central.

7.6 Which group matters most?

White women account for most of the gains because they are a large share of the population. In Table VII, falling barriers for white women account for 33.8% of market GDP per person growth, compared with 1.2% for black men and 3.7% for black women.

7.7 Residual gains after 2010

The paper asks how much additional GDP could be achieved by removing remaining 2010 frictions. The answer is 9.9% higher GDP. This is meaningful, but much smaller than the historical gains from 1960 to 2010. The paper therefore suggests that a large part of the growth gain from occupational convergence has already occurred.

8 Robustness and alternative interpretations

8.1 Parameter robustness

The baseline result is robust to varying θ , η , and σ . The most sensitive parameter is σ , the elasticity of substitution across occupations. With $\sigma = 1.05$, the share of market GDP per person growth explained by barriers falls to 22.2%, but this mainly reallocates growth between market and home sectors. The total GDP gains are more stable.

8.2 Preference sorting

The paper allows a fraction δ of workers to sort on idiosyncratic preferences rather than talent. The authors estimate $\delta = 0.22$. With this estimate, falling barriers still explain 37.4% of market GDP per person growth. Even with $\delta = 0.5$, the effect is 28.3%. In the polar case $\delta = 1$, where all sorting is by preferences, τ^h and τ^w alone explain only 5.6%, but τ^h , τ^w , and $\tilde{z}$ together still explain 19.8%.

8.3 Alternative assumptions about talent

The paper tests whether the results are an artifact of assuming identical talent across groups. Two exercises are especially important:

- In “brawny” occupations such as construction, manufacturing, police, and firefighters, the authors allow group talent differences to explain quantity differences. The market GDP contribution falls only from 41.5% to 38.7%.
- If all group differences in 2010 are assumed to reflect innate talent rather than frictions, the contribution of falling frictions in earlier years remains 35.8%.

The reason is that most gains come from women entering high-skilled occupations such as law, medicine, science, academia, and management, where physical strength explanations are not persuasive.

8.4 Why reverse-discrimination stories are hard to fit

The paper discusses alternatives in which men became increasingly disadvantaged or women became increasingly favored. These stories do not fit the data well. If women and men had identical talent and there were no discrimination in 1960, the model would predict similar wages and occupational distributions in 1960, which is not observed. If women were innately less talented and later received favorable discrimination, the model would imply rising misallocation, but AFQT evidence does not support lower female talent.

9 Limitations and interpretation

- **Identification is structural.** The results rely on the assumption that relative group talent in occupations is stable over time. This is plausible for many high-skilled occupations, but it cannot be proven directly.
- **Barriers are broad residuals.** Human capital barriers include schooling, socialization, parental investment, wealth constraints, access to professional training, and other pre-market forces. Labor market barriers include multiple forms of discrimination and wage wedges.
- **Preferences versus barriers are hard to separate.** Social norms can look like preferences in the model, but they may themselves be barriers. The paper is careful to show that the growth effects are large even when this interpretation varies.
- **No direct welfare calculation.** The paper quantifies GDP and productivity, not welfare from home production, occupational preferences, or changes in family structure.
- **Model tractability.** The Fréchet distribution and the absence of correlation between absolute and comparative advantage make the model tractable but are simplifying assumptions.

10 Tables and figures

10.1 Figure 1: Standard deviation of relative occupational shares and Figure 2: Mean composite occupational frictions

- **Figure 1 read.** The standard deviation of relative occupational shares falls for white women and black women, and also falls for black men. The most dramatic reductions occur before 1980.
- **Figure 2 read.** Mean composite frictions fall sharply for all three non-white-male groups, especially for women. The decline is largest from 1960 to 1980.

- **Economic message.** The raw occupational convergence and the model-implied friction decline are the first pieces of evidence that barriers to occupational choice became much smaller over the period.

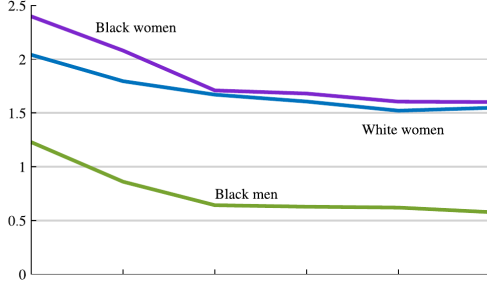


Figure 1: Standard deviation of relative occupational shares

Occupational distributions of women and black men converge toward that of white men.

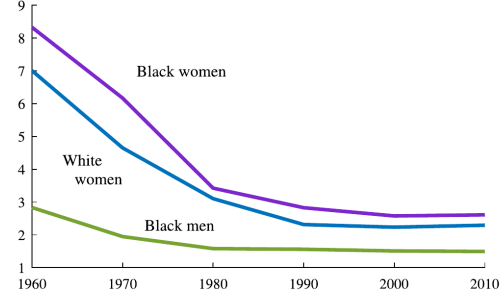


Figure 2: Mean composite occupational frictions

The model infers large declines in the mean composite barrier τ for white women, black men, and black women.

10.2 Figure 3: Variance of composite frictions and preferences and Figure 4: Occupational barriers for white women

- **Figure 3 read.** The variance of $\log \tau$ declines for all three groups, while the variance of occupation-specific preferences $\log \tilde{z}$ changes much less.
- **Figure 4 read.** For white women, barriers were very high in 1960 for lawyers, doctors, and construction. Barriers to law and medicine decline dramatically; barriers in construction remain high.
- **Economic message.** Most occupational convergence is attributed to changing barriers, not changing tastes. The growth-relevant reductions are concentrated in high-skilled occupations.

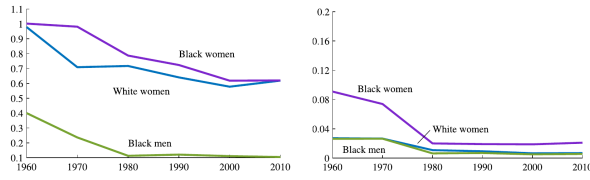


Figure 3: Variance of composite frictions and preferences

The dispersion of frictions falls much more than the dispersion of preferences.

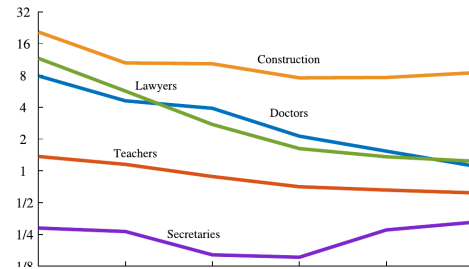


Figure 4: Occupational barriers for white women

White women's barriers fall sharply in law and medicine, but not in construction.

10.3 Figure 5: Wage gaps relative to white men by time and cohort, and Figure 6: Mean occupational preferences

- **Figure 5 read.** Wage gaps relative to white men are plotted by cohort and age. White women's profiles shift upward across cohorts, consistent with lower human capital barriers. Black men's profiles show both intercept and slope changes.
- **Figure 6 read.** Mean occupational preference for market work relative to white men rises for white women and black women and falls for black men.
- **Economic message.** The life-cycle structure is what allows the paper to distinguish pre-market human capital barriers from contemporaneous labor-market wedges.

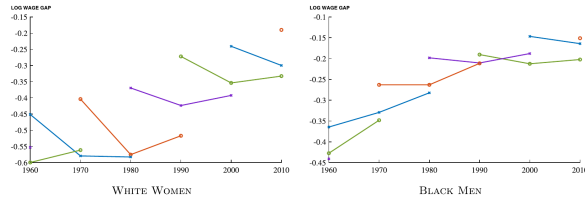


Figure 5: Wage gaps relative to white men by time and cohort

Cohort shifts versus within-cohort slopes help separate human capital barriers from labor market wedges.

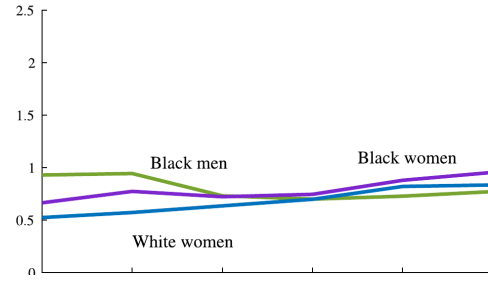


Figure 6: Mean occupational preferences
The model infers rising white women's preference for market work relative to home production.

10.4 Table I: Identifying assumptions and normalizations, and Table II: Baseline parameter values

- **Table I read.** The baseline normalizes white men's labor and human capital frictions to zero, home-sector frictions to zero for all groups, home-sector preference to one, and relative talent to one.
- **Table II read.** The baseline calibration uses $\theta = 2$, $\eta = 0.103$, $\sigma = 3$, and $\beta = 0.231$.
- **Economic message.** These normalizations make the barriers relative to white men. The paper does not identify absolute discrimination levels; it identifies changes in relative barriers over time.

TABLE I
IDENTIFYING ASSUMPTIONS AND NORMALIZATIONS

Parameter	Definition	Determination	Value
τ_{wm}^h	Human capital barriers (white men)	Assumption	0
τ_{wm}^l	Labor market barriers (white men)	Assumption	0
$\bar{h}_{i,g}$	Talent in each occupation (all groups)	Assumption	1
$\tau_{home,g}^h$	Home human capital barriers (all groups)	Assumption	0
$\tau_{home,g}^l$	Home labor market barriers (all groups)	Assumption	0
$\bar{z}_{home,g}$	Home occupational preference (all groups)	Normalization	1

Table I: Identifying assumptions and normalizations

The baseline assumes no white-male barriers and equal group talent in all occupations.

TABLE II
BASELINE PARAMETER VALUES

Parameter	Definition	Determination	Value
θ	Fréchet shape	Wage dispersion, Frisch elasticity	2
η	Goods elasticity of human capital	Education spending	0.103
σ	EoS across occupations	Arbitrary	3
β	Consumption weight in utility	Mincerian return to education	0.231

Table II: Baseline parameter values

The main parameters are chosen from wage dispersion, education spending, and Mincerian returns.

10.5 Table III: Forcing variables and empirical targets when $\delta = 0$ and Table IV: Model versus data

- **Table III read.** The model chooses technology, returns to schooling, human capital barriers, labor market barriers, preferences, and experience terms to match occupational shares, education, life-cycle wage growth, wages, and white men’s age-earnings profile.
- **Table IV read.** The model matches aggregate earnings and labor force participation reasonably well despite targeting mainly the young’s occupational shares and participation.
- **Economic message.** The model is disciplined by multiple moments, not only by aggregate GDP. This makes the growth counterfactual more credible.

TABLE III
FORCING VARIABLES AND EMPIRICAL TARGETS WHEN $\delta = 0^a$

Parameter	Definition	Empirical Target
$A_i(t)$	Technology by occupation	Occupations of young white men
$\phi_y(t)$	Time elasticity of human capital	Education by occupation, young white men
$\tau_{ig}^h(t)$	Human capital barriers	Occupations of the young, by group
$\tau_{ig}^l(t)$	Labor market barriers	Life-cycle wage growth, by group
$\bar{z}_{ig}(c)$	Occupational preferences	Wages by occupation for the young
$\gamma(1), \gamma(2)$	Experience terms	Age earnings profile of white men

^aThe variable values are chosen jointly to match the empirical targets. $\delta = 0$ refers to the polar case where individuals draw idiosyncratic ability for each occupation (but not idiosyncratic tastes).

Table III: Forcing variables and empirical targets when $\delta = 0$

The model uses different empirical moments to infer each forcing variable.

TABLE IV
MODEL VERSUS DATA: EARNINGS AND LABOR FORCE PARTICIPATION^a

Year	Earnings Data	Earnings Model	LFP Data	LFP Model
1960	18,383	18,615	0.599	0.599
1970	24,645	25,000	0.636	0.614
1980	27,088	27,900	0.702	0.653
1990	33,953	34,265	0.764	0.720
2000	39,419	41,134	0.747	0.743
2010	41,541	42,717	0.759	0.748

^aThis table shows average market earnings per worker in 2009 dollars and labor force participation in the Census/ACS data alongside the corresponding model values by year.

Table IV: Model versus data

The model closely tracks aggregate earnings per worker and labor force participation.

10.6 Table V: Share of growth due to changing frictions and Figure 7: GDP per person, data and counterfactual

- **Table V read.** Falling human capital and labor market frictions explain 41.5% of market GDP per person growth and 90.4% of labor force participation growth. Human capital barriers alone explain 36.0%; labor market barriers alone explain 7.7%.
- **Figure 7 read.** The counterfactual with 1960 barriers grows much less than the model/data path, especially after 1980.
- **Economic message.** The gains from occupational convergence are large. Without the reduction in barriers, the paper’s model implies substantially slower growth in market output per person.

	Share of growth accounted for by			
	τ^h and τ^w	$\tau^h, \tau^w, \bar{z}$	τ^h only	τ^w only
Market GDP per person	41.5%	40.8%	36.0%	7.7%
Market earnings per person	38.4%	37.5%	18.9%	26.0%
Labor force participation	90.4%	112.7%	24.9%	56.2%
Market GDP per worker	24.0%	15.0%	40.0%	-9.8%
Home + market GDP per person	32.7%	32.1%	30.6%	4.4%

^aEntries in the table show the share of growth in the model attributable to changing frictions under various assumptions. The variables are τ^h (human capital frictions), τ^w (labor market frictions), and $\bar{z}$ (occupational preferences).

Table V: Share of growth due to changing frictions

Declining barriers account for roughly two-fifths of market GDP per person growth.

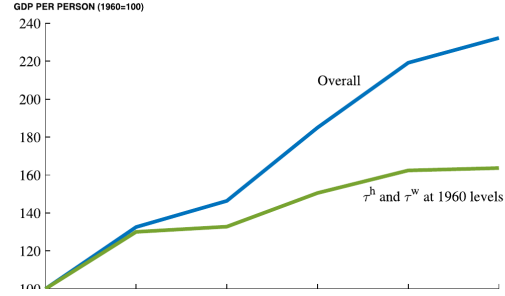


Figure 7: GDP per person, data and counterfactual

Holding frictions fixed at 1960 levels produces much lower cumulative market GDP growth.

10.7 Table VI: Wage gaps and earnings by group and Table VII: Share of growth due to different groups

- **Table VI read.** Falling barriers explain most of the rising earnings of women and black men and more than explain the decline in white women's wage gap relative to white men.
- **Table VII read.** White women account for most of the aggregate growth contribution: 33.8 percentage points out of the 41.5% total contribution.
- **Economic message.** The largest aggregate gains come from a large population group moving into higher-productivity occupations.

TABLE VI
WAGE GAPS AND EARNINGS BY GROUP AND CHANGING FRICTIONS^a

	Share of growth accounted for by		Full Model
	τ^h and τ^w	$\tau^h, \tau^w, \bar{z}$	
Earnings, WM	-12.2%	-17.4%	91.4%
Earnings, WW	76.9%	86.3%	107.0%
Earnings, BM	28.7%	20.4%	100.5%
Earnings, BW	50.6%	52.5%	101.7%
Wage gap, WW	148.1%	98.3%	176.7%
Wage gap, BM	98.0%	115.4%	143.4%
Wage gap, BW	84.6%	71.4%	131.4%
LF Participation	90.4%	112.7%	93.6%

^aEntries in the table show the share of growth in the model attributable to changing frictions and other variables. The frictions are τ^h (human capital) and τ^w (labor market), and $\bar{z}$ are occupational preferences. The last column reports the share of observed growth explained by the full model solution, including the A and ϕ variables.

Table VI: Wage gaps and earnings by group
The model attributes much of group-specific earnings convergence to falling barriers.

TABLE VII
SHARE OF GROWTH IN MARKET GDP PER PERSON DUE TO DIFFERENT GROUPS^a

1960-2010	τ^h and τ^w	τ^h only	τ^w only
All groups	41.5%	36.0%	7.7%
White women	33.8%	29.8%	6.1%
Black men	1.2%	0.7%	0.6%
Black women	3.7%	3.2%	0.9%

^aEntries are the share of growth in GDP per person from changing frictions for various groups over different time periods. The variables are τ^h (human capital barriers), and τ^w (labor market frictions).

Table VII: Share of growth due to different groups

Most aggregate gains come from reductions in barriers facing white women.

10.8 Table VIII: Robustness to parameter values and Table IX: Forcing variables and empirical targets when $\delta = 1$

- **Table VIII read.** The baseline growth contribution remains large under alternative values of θ , η , and σ . The market-output effect is smaller when occupations are close to Cobb-Douglas substitutes.
- **Table IX read.** When workers sort only on idiosyncratic preferences, the empirical targets change: wages by occupation identify human capital barriers, and occupational shares identify preferences.
- **Economic message.** The paper’s qualitative conclusion is not driven by a single parameter choice or by the exact talent-versus-preference interpretation.

TABLE VIII
ROBUSTNESS TO PARAMETER VALUES^a

	Growth in GDP per person accounted for by		
	τ^h and τ^w	τ^h alone	τ^w alone
Benchmark	41.5%	36.0%	7.7%
$\theta = 1.5$	45.3%	41.2%	6.8%
$\theta = 4$	32.2%	25.2%	7.0%
$\eta = 0.05$	39.8%	33.8%	8.2%
$\eta = 0.20$	44.4%	39.4%	7.2%
$\sigma = 1.05$	22.2%	19.8%	3.6%
$\sigma = 10$	42.9%	36.5%	8.5%

^aEntries show the share of growth in the model attributable to changing frictions τ^h (human capital) and τ^w (labor market). The benchmark parameter values are $\theta = 2$, $\eta = 0.103$, and $\sigma = 3$.

Table VIII: Robustness to parameter values
The market GDP contribution remains sizable across
alternative parameter calibrations.

TABLE IX
FORCING VARIABLES AND EMPIRICAL TARGETS WITH $\delta = 1$ ^a

Parameter	Definition	Empirical Target
$A_i(t)$	Technology by occupation	Occupations of young white men
$\phi_i(t)$	Time elasticity of human capital	Education by occupation, young white men
$\tau_{\theta}^h(c)$	Human capital barriers	<i>Wages by occupation for the young</i>
$\tau_{\theta}^w(t)$	Labor market barriers	Life-cycle wage growth, by group
$\bar{z}_{ig}(c)$	Occupational preferences	<i>Occupations of the young, by group</i>
$\gamma(1), \gamma(2)$	Experience terms	Age earnings profile of white men

^aVariables are chosen to match the empirical targets. $\delta = 1$ is the polar case where individuals draw idiosyncratic tastes for each occupation, with no idiosyncratic heterogeneity in ability across occupations.

Table IX: Forcing variables and empirical targets
when $\delta = 1$
The polar preference-sorting case uses different moments
to infer barriers and preferences.

10.9 Figure 8: Wage gaps versus propensities, white women in 1980, and Figure 9: Changes in wage gaps versus propensity gaps

- **Figure 8 read.** For young white women in 1980, there is little systematic relationship between occupational wage gaps and relative propensities across occupations.
- **Figure 9 read.** From 1960 to 2010, changes in wage gaps and changes in propensity gaps for young white women are also weakly related.
- **Economic message.** These weak relationships are more consistent with sorting on talent than with pure preference sorting, because preference sorting would predict stronger links between wage compensation and occupational shares.

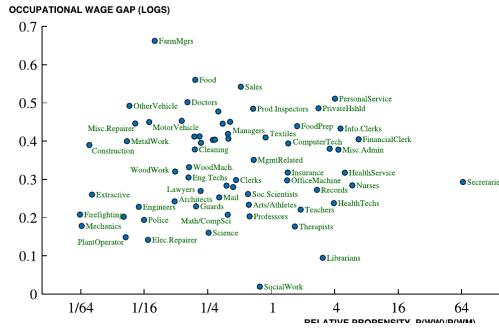


Figure 8: Wage gaps versus propensities, white women in 1980

The cross-occupation relationship is weak, undermining a pure preference-sorting interpretation.

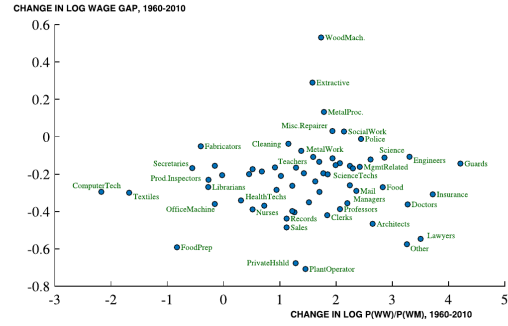


Figure 9: Changes in wage gaps versus propensity gaps

The change relationship over 1960–2010 is also weak.

10.10 Figure 10: Changes in wages versus propensities, white men, and Table X: Allowing for selection on idiosyncratic preferences

- **Figure 10 read.** Changes in schooling-adjusted wages and relative propensities for young white men are used to infer a plausible value of δ .
- **Table X read.** With $\delta = 0.22$, falling human capital and labor market barriers explain 37.4% of market GDP per person growth, close to the 41.5% baseline. With $\delta = 1$, barriers alone explain much less, but barriers plus preferences still explain 19.8%.
- **Economic message.** The exact split between barriers and social norms matters less for the conclusion that occupational convergence made a large contribution to growth.

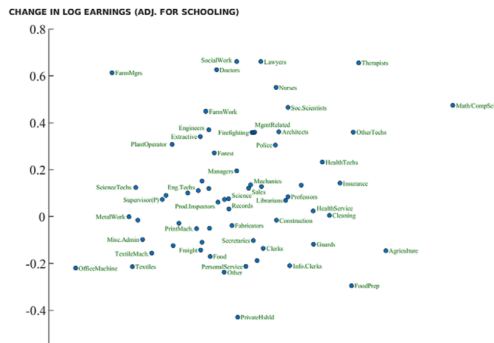


Figure 10: Changes in wages versus propensities, white men

The authors use this relationship to calibrate the share of sorting due to preferences.

TABLE X
ALLOWING FOR SELECTION ON IDIOSYNCRATIC PREFERENCES^a

	Share of growth accounted for by τ^h and τ^w			
	$\delta = 0$	$\delta = 0.22$	$\delta = 0.5$	$\delta = 1$
Market GDP per person	41.5%	37.4%	28.3%	5.6%
Market earnings per person	38.4%	37.0%	31.2%	11.5%
Labor force participation	90.4%	88.1%	74.9%	21.2%
Market GDP per worker	24.0%	20.6%	14.2%	0.1%
Home + market GDP per person	32.7%	31.3%	25.7%	1.6%

	Share of growth accounted for by τ^h , τ^w , and ξ			
	$\delta = 0$	$\delta = 0.22$	$\delta = 0.5$	$\delta = 1$
Market GDP per person	40.8%	38.9%	33.7%	19.8%
Market earnings per person	37.5%	36.8%	33.4%	20.8%
Labor force participation	112.7%	117.6%	117.1%	71.2%
Market GDP per worker	15.0%	12.7%	8.6%	1.8%
Home + market GDP per person	32.1%	32.3%	29.8%	7.5%

^a Entries in the table show the share of growth in the model attributable to changing frictions τ^h (human capital frictions), τ^w (labor market frictions), and ξ (occupational preferences), for chosen values of δ , the fraction of workers who show occupational convergence.

Table X: Allowing for selection on idiosyncratic preferences

Growth contributions remain large for plausible δ , and remain economically meaningful even in the polar preference case.

10.11 Table XI: Robustness to alternative assumptions about group differences in talent

- **Table XI read.** Allowing gender-specific talent in brawny occupations lowers the contribution only modestly, from 41.5% to 38.7%. Assuming all 2010 group differences reflect talent still leaves a 35.8% contribution from earlier falling barriers.
- **Economic message.** The results do not rest on assuming that women and men have identical physical ability in every occupation. The key gains come from high-skilled occupations where physical-strength explanations are less relevant.

TABLE XI
ROBUSTNESS TO ALTERNATIVE ASSUMPTIONS ABOUT GROUP DIFFERENCES
IN TALENT^a

	Market GDP per person growth accounted for by τ^h and τ^w
Benchmark	41.5%
No frictions in “brawny” occupations	38.7%
No frictions in 2010	35.8%

Table XI: Robustness to alternative assumptions about group differences in talent

11 Conclusion

- **Core conclusion.** The improved allocation of talent across occupations explains a substantial share of U.S. growth from 1960 to 2010. In the baseline, falling barriers explain roughly two-fifths of market GDP per person growth.
- **Most important channel.** Declining barriers to human capital accumulation are more important for output growth than declining labor market discrimination. The interpretation is that opening access to skills, education, and professional preparation mattered enormously.
- **Role of preferences.** Social norms and preferences help explain labor force participation and occupational sorting, but in the baseline they explain little market GDP growth. Under broader preference-sorting assumptions, the combined effect of changing barriers and preferences remains economically important.
- **Distributional implication.** Falling barriers raised the earnings of women and black men, while also reallocating white men away from some high-skilled occupations. This general-equilibrium response is part of the paper’s accounting.
- **Growth implication.** Reducing discrimination and barriers is not only an equity issue. It is also an aggregate productivity issue because it changes who uses scarce talent in high-return occupations.
- **Takeaway.** The paper reframes occupational inclusion as a major source of economic growth: when more people are allowed to pursue their comparative advantage, the economy becomes substantially more productive.